

Supplementary Material

This supplementary material lists implementation details used in the GrapeMaster platform. The main manuscript reports the platform architecture, data flow, analytical modules, validation design, and workflow verification; this document provides supplementary linkage figures, database relationship rules, workflow verification rules, the software stack, selected data-model attributes, and operational-record screening rules for reproducibility and technical inspection.

Supplementary figures

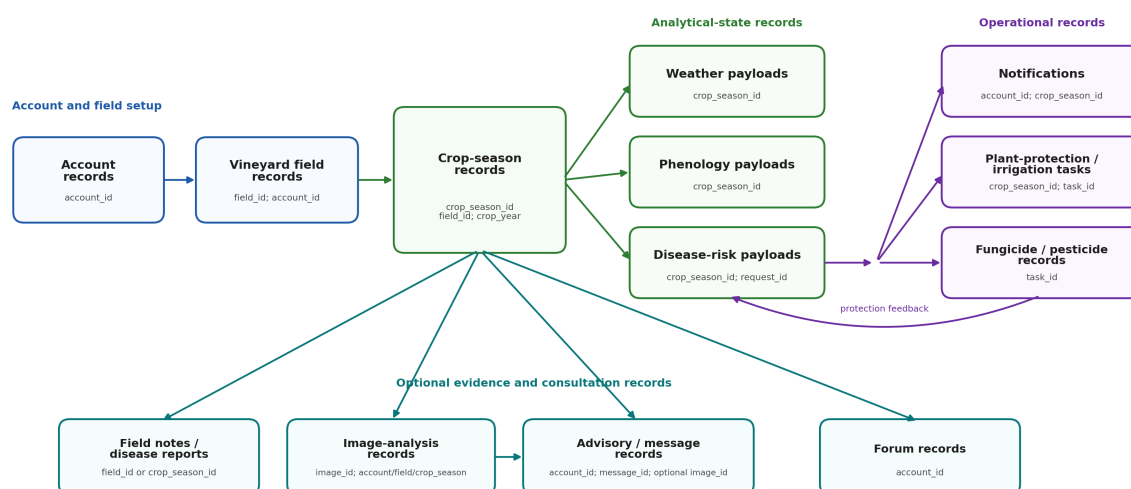


Figure S1. Backend identifier linkage among GrapeMaster record classes. The crop-management record is shown as a data-linkage object connecting field setup with time-dependent analytical and operational records. Task identifiers directly connect field operations with product-level treatment feedback; image and advisory records use the available account, field, crop, image, or message context. The figure is a schematic linkage view and does not report deployment counts or causal relationships.

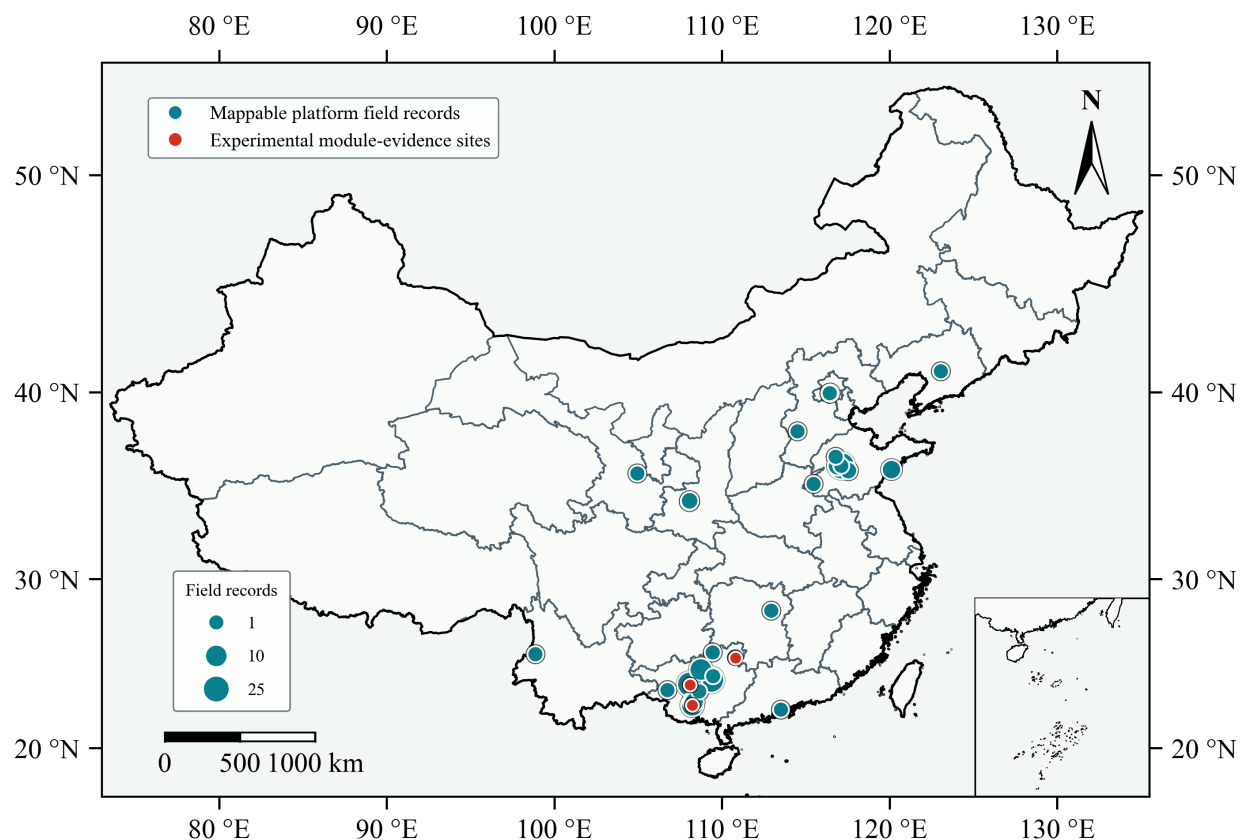


Figure S2. Spatial context of mappable platform field records and independent experimental module-validation sites. Teal bubbles show mappable field records aggregated by region, with bubble size indicating the number of records per region. Red points indicate experimental sites used for module-level validation and are not part of the operational-record denominator.

Supplementary tables

Table S1. Input-output definitions and calibration notes for platform-compatible analytical components.

Module	Inputs and outputs	Workflow position and calibration note
Phenology service	Inputs include crop-season metadata, cultivar and cultivation context, and date-aligned daily temperature or weather series. Outputs include broad BBCH-related growth-stage states, stage timelines, and daily crop-stage sequences.	Provides host-development context for susceptibility interpretation, warning-window definition, task timing, and mobile display. The implementation is a Guangxi-calibrated adaptation of temperature-driven grapevine phenology modelling; parameter thresholds and validation metrics are retained as module-level configuration.
FSIM-S risk service	Inputs include transformed hourly or daily weather, crop-stage sequence, cultivar susceptibility, target-disease settings, and recent fungicide-history records. Outputs include downy mildew disease-risk states, first-infection warning triggers, treatment-window information, and protection-state interpretation.	Supplies pre-symptomatic risk information to the proactive management workflow. The evaluated configuration is locally calibrated for humid subtropical Guangxi vineyards. Field-derived sporangia pressure is used as an optional model-level calibration or sensor-enhanced pathway, not as a required live input for routine operation.
Image-recognition service	Inputs include user-uploaded grape symptom images. Outputs include disease or healthy categories, confidence values, and stored image-analysis records.	Supplies post-symptomatic diagnostic evidence to the mobile application. Candidate visual models were evaluated with grape-focused validation metrics, including accuracy, precision, recall, and F1-score.

Module	Inputs and outputs	Workflow position and calibration note
LLM-assisted advisory service	Inputs include advisory text and available diagnostic or management context. Outputs include management-oriented advisory text and stored advisory records.	Supplies contextual explanation and conversational advisory support after image recognition or user questioning. The service follows the CDIP-ChatGLM3 framework and is not used as a substitute for validated disease models, expert diagnosis, pesticide labels, or local regulations.

Table S2. Database relationship rules supporting model-to-management workflow reconstruction.

Relationship rule	Required linkage	Verification purpose
Account-to-field linkage	Retained account identifiers must link to at least one vineyard field record	Confirms that the account has an operational vineyard context rather than registration-only activity
Field-to-crop linkage	Each evaluated crop-management record must be linked to a field record through field or crop identifiers	Separates persistent field identity from time-dependent analytical and management state
Crop-to-analytical-payload linkage	Weather, phenology, disease-risk, and model request-response payloads should share the crop identifier where applicable	Confirms that model inputs and outputs can be retrieved under the same management context
Context-to-event linkage	Notifications, warning events, and task records should be associated with the retained user, field, or crop context	Confirms contextual association without implying direct event-to-task causality
Task-to-product linkage	Fungicide or pesticide product records must link to plant-protection tasks through task identifiers	Confirms that completed operations can be converted into product-level treatment feedback

Relationship rule	Required linkage	Verification purpose
Symptom/advisory context linkage	Image records, advisory records, field notes, and disease reports should retain available account, field, crop, image, or message identifiers	Confirms whether symptom-related records and advisory interactions can be retrieved with relevant management information
Explicit absence rule	If an expected optional record type is absent for a replay anchor, the absence is recorded rather than inferred as failure of unrelated workflow steps	Prevents overclaiming and separates missing symptom or advisory records from successful risk-task-feedback linkage

Table S3. Workflow verification rules used for operational-record and representative-replay checks.

Verification check	Operational criterion	Supported interpretation
Operational-account retention	An account is retained only when it has business records and field-crop-season linkage after deduplication and target-context screening	Defines the evaluated operational account set
Analytical payload completeness	A crop season is counted as analytically complete when weather, phenology, disease-risk, and request-response payloads are present under the shared seasonal context	Shows whether the analytical state can be reconstructed for that season
Disease-risk-to-operation linkage	Notification and task records are checked against retained user, field, or crop-season context	Shows whether disease-risk or management events entered the operational workflow
Treatment-feedback linkage	Fungicide or pesticide records are checked through task identifiers and task completion status	Shows whether field operations can be converted into feedback records
Symptom/advisory record-linkage availability	Image-analysis, advisory, disease-report, and note records are checked for available account, field, crop-season, image, or message links	Shows whether symptom-related records and advisory interactions are available for seasonal interpretation

Verification check	Operational criterion	Supported interpretation
Representative replay	A selected crop-season anchor must retrieve weather, phenology, disease-risk, notification, task, and fungicide records through shared crop-season or task keys	Demonstrates reconstructability of one disease-risk-to-feedback chain
Boundary reporting	Missing optional symptom or advisory records under the replay anchor are explicitly reported rather than omitted	Separates verified linkage from unsupported production-outcome claims
Excluded outcome claim	Operational-record linkage and replay checks are not treated as demonstrations of pesticide reduction, disease-control efficacy, economic benefit, or user adoption	Maintains the distinction between record-based workflow assessment and agronomic-impact evaluation

Table S4. Record-level reconstruction results for a loop-complete crop-season replay in GrapeMaster.

Record group	Retained record set	Identifier or status	Result note
Replay record set	Weather, phenology, disease-risk, notification, task, and fungicide records	One crop-season UUID	Loop-complete crop-season replay
State and risk records	Weather, phenology, and disease-risk payloads	Shared crop-season key	Shared seasonal record set
Operation records	Notifications and completed plant-protection tasks	Same crop-season context	Notification and task records
Treatment-feedback records	Fungicide-product records	Task identifiers	Product-task record set
Field-note branch	No field-note or incidence-note record	Explicit absence	No field/incidence note in this replay

Table S5. Software stack and implementation components used in the GrapeMaster platform.

Layer	Component group	Main technologies or libraries
Mobile client	Application framework	Dart, Flutter
Mobile client	Android shell and build configuration	Kotlin, Gradle
Backend business service	Web framework and API layer	Python, Django 4.2, Django REST Framework
Backend business service	Authentication, scheduling, and asynchronous communication	SimpleJWT, Django APScheduler, Django Channels
Backend business service	Database access and persistence	PostgreSQL, psycopg2, normalized relational tables, JSON model payloads
Disease-model service	Service interface	FastAPI, Uvicorn, Pydantic, Starlette
Weather and time-series processing	Numerical and date-time processing	NumPy, Pandas, Python Dateutil, Pytz
Weather and geospatial processing	Field geometry, geocoding, and configured weather-service access	Shapely, Fiona, GeoPandas, Geopy, Requests, HTTPX, weather-service client packages, request caching, retry utilities
Phenology and risk analytics	Crop-stage and risk-state calculation	Python analytical workflows, date-aligned weather transformation, cultivar and crop-season metadata
Image processing and recognition	Image preprocessing and deep-learning inference	PyTorch, TorchVision, OpenCV, Pillow, Ultralytics-related components, EfficientViT-related modules, segmentation components
Advisory interface	Text advisory channel	ChatGLM3-derived advisory workflow following the published CDIP-ChatGLM3 framework

Table S6. Selected data-model attributes supporting model-to-management integration.

Record group	Main attributes or stored content	Platform role
Field record	User linkage, field name, area, GeoJSON boundary, centroid coordinates, administrative region, sharing state, deletion state, notice state, and generated field-boundary preview	Persistent spatial and ownership unit for vineyard management
Field geometry processing	Boundary parsing, boundary-image generation, area estimation, centroid calculation, and optional geocoding request	Converts user-defined field boundaries into location and display objects
Crop-management record	Field linkage, grape cultivar, cultivation method, vine age, field coordinates, field size, and cultivar metadata	Time-dependent execution and retrieval key for analytical and operational records
Master-data records	Grape variety, cultivar susceptibility, cultivation method, BBCH stage, disease type, fungicide, pesticide, and reminder rules	Controlled reference values for field, crop, task, risk, and advisory records
Model payload records	Weather, phenology, disease-risk, and request-response payloads stored with structured records	Preserves inputs and outputs for mobile display, recalculation, and audit

Table S7. Operational-record screening rules used before summarizing deployed GrapeMaster records.

Screening step	Rule	Rationale
User-record integration	User records were integrated using available user identifier, custom-user identifier, and phone-number fields.	Avoids duplicate counting of account-level records

Screening step	Rule	Rationale
Business-record filter	Accounts with no business records across field, crop-season, notification, task, fungicide or pesticide, advisory or message, forum, disease-report, and note tables were excluded.	Removes registration-only accounts from operational summaries
Field-crop linkage filter	Accounts without both field records and crop-management records were excluded.	Retains records that can support implemented workflow reconstruction
Target-context filter	A single account associated only with Linwei District was excluded.	Removes a non-target-region account from the Guangxi vineyard evaluation context
Retained account set	Forty-seven accounts remained from 95 deduplicated accounts.	Defines the candidate account set used for operational-record coverage